Healthcare Operations & Billing

Business Insights from 54,966 Patient Records

Novexa Technologies — Data Analysis Task 9

Executive Summary

- Billing is remarkably uniform across medical conditions — varying only 2-3% around a $25,546 average
- Length of stay is nearly identical across all six conditions (~15.4-15.7 days)
- Patient volume splits almost evenly across Elective, Urgent, and Emergency admissions
- A data quality issue was found and corrected: 108 records had negative billing values

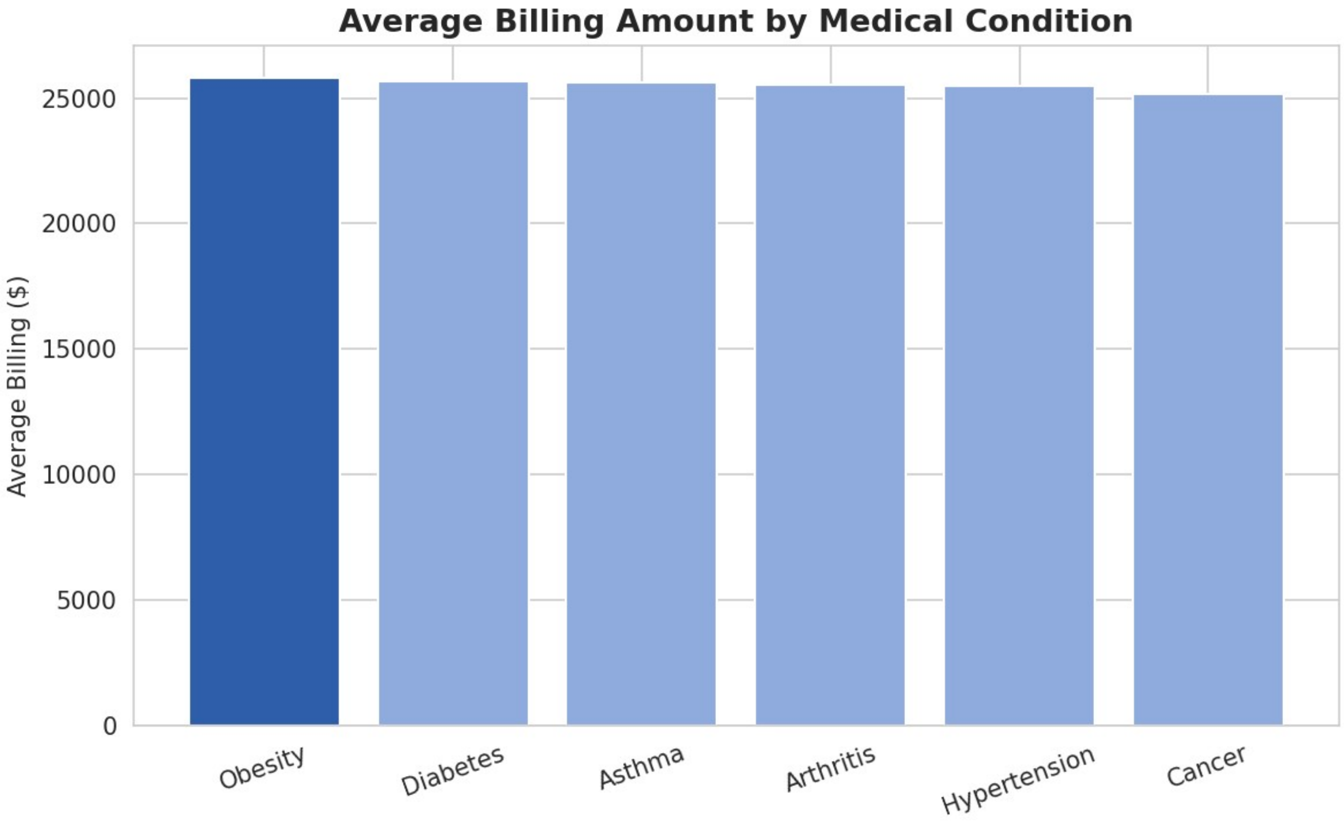
The Dataset at a Glance

54,966	**$1.40B**	**$25,546**	**15.5 days**
Patient Records (after removing 534 duplicates)	Total Billed Revenue	Average Billing per Patient	Average Length of Stay

Age range 13-89 (avg 51.5) • Gender split near 50/50 • 6 medical conditions • 5 insurance providers

Finding 1
Billing Doesn't Scale With Condition Complexity

Average Billing by Medical Condition

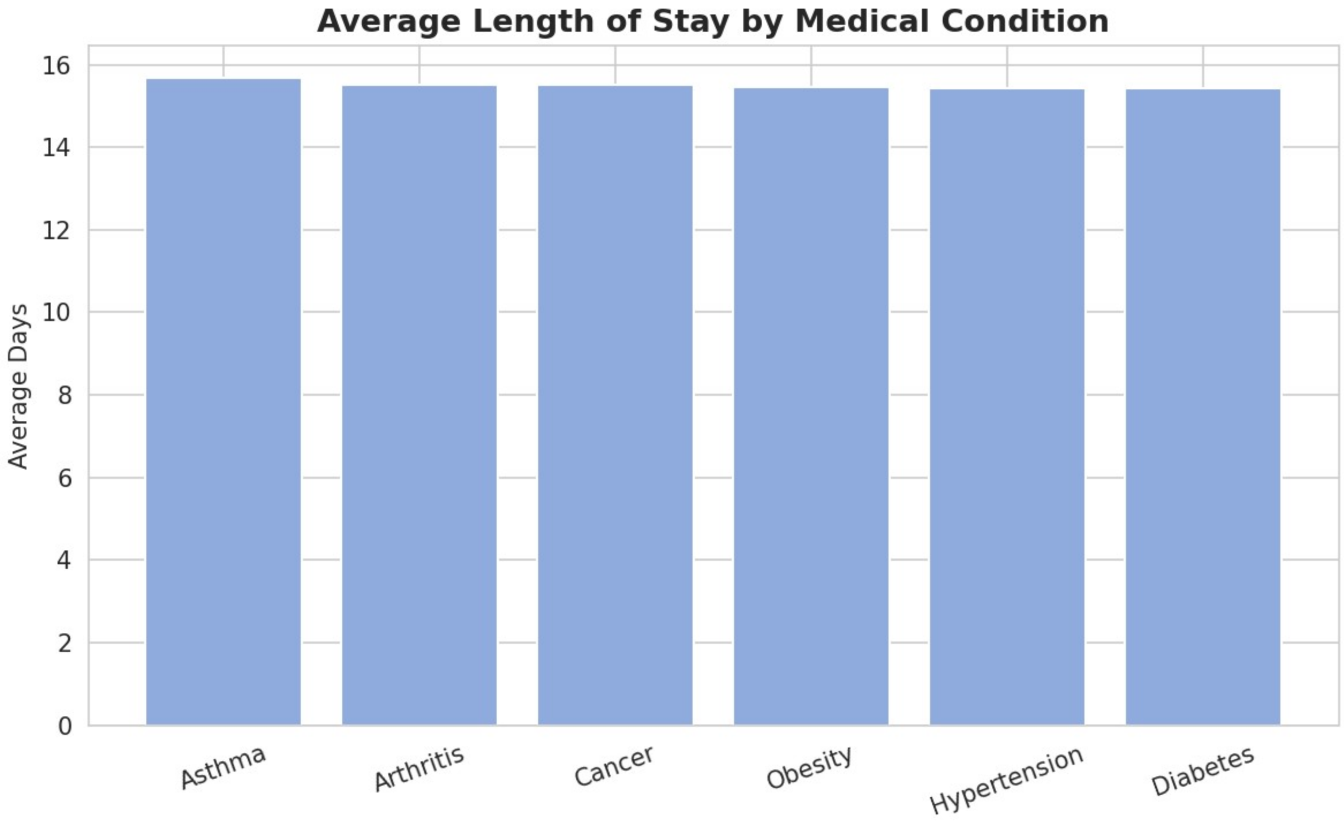


Takeaway

Obesity is highest ($25,807) and Cancer is lowest ($25,155) — only a 2.6% spread. This suggests flat-rate or bundled pricing rather than condition-driven cost. Worth confirming whether this is intentional.

Finding 2
Length of Stay Is Nearly Identical Across Conditions

Average Length of Stay by Condition

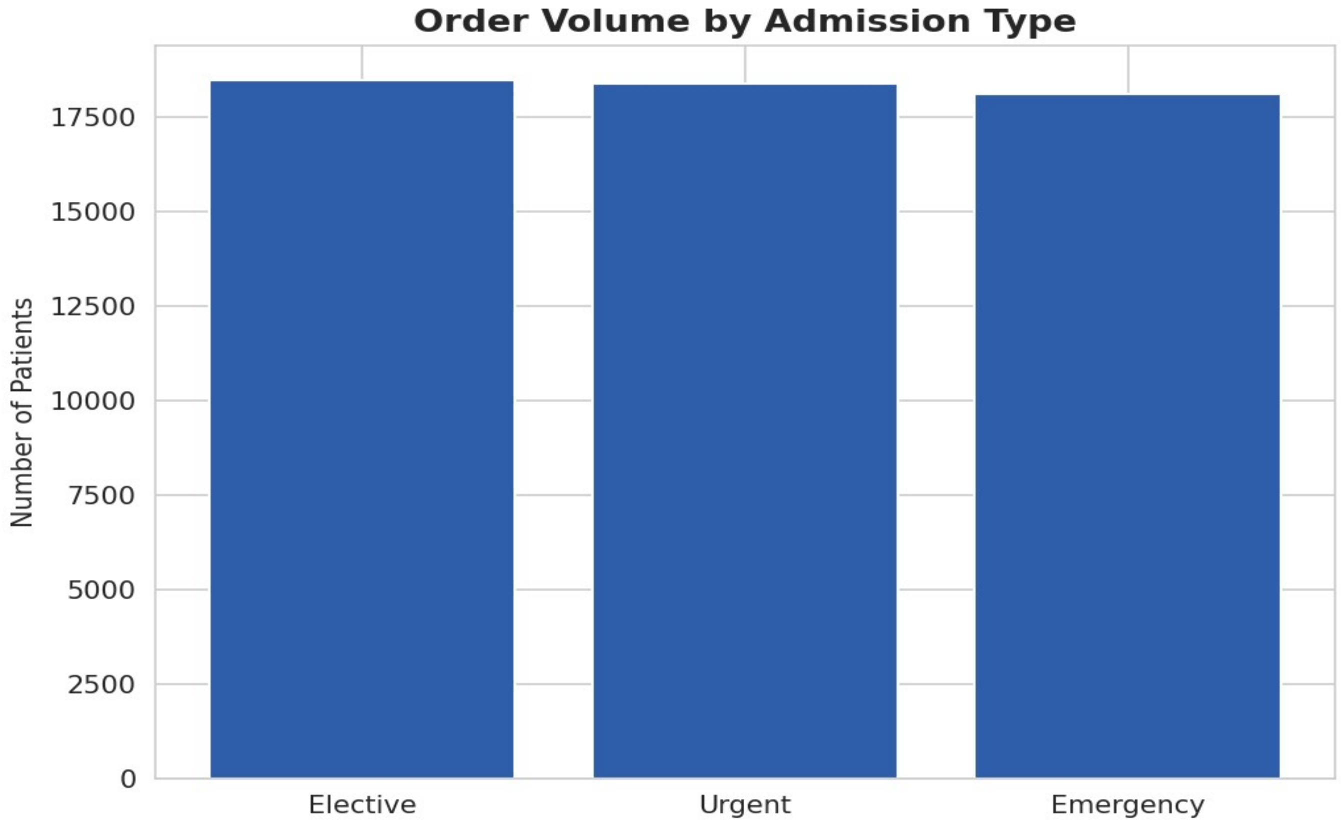


Takeaway

All six conditions fall within 15.4-15.7 days. This points to a standardized care pathway, and simplifies bed-capacity planning since stay length doesn't need to be modeled per condition.

Finding 3
Admissions & Outcomes Split Evenly Three Ways

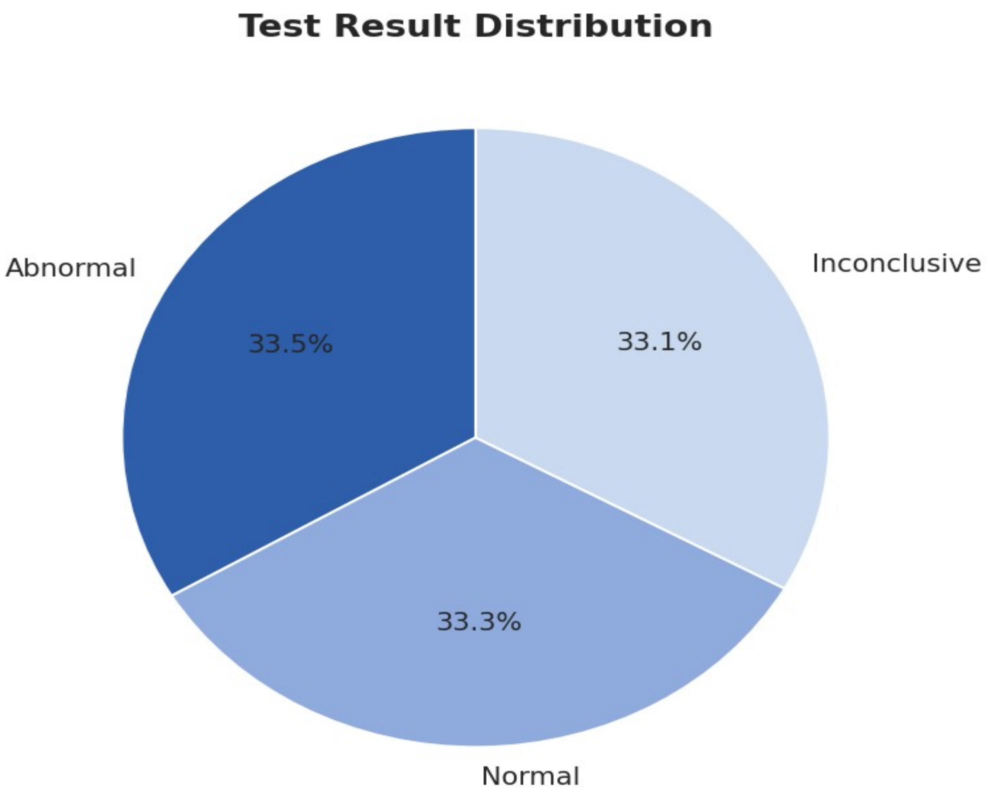
Admission Type Volume



Takeaway

Elective, Urgent, and Emergency each account for roughly a third of admissions. Emergency capacity should be planned for ~33% of total demand, not treated as an edge case.

Test Result Distribution



Takeaway

Abnormal, Normal, and Inconclusive results are each close to a third of all tests — no single outcome dominates.

Recommendations

1. Investigate why billing doesn't scale with condition complexity — confirm bundled pricing vs. a data gap
2. Use uniform length of stay as a bed-capacity planning baseline
3. Plan emergency/urgent capacity for ~1/3 of total patient volume
4. Add a standing check for negative billing values in future data exports

Questions?

Full analysis, code, and dataset available in the project repository